

CS 57800 – Statistical Machine Learning

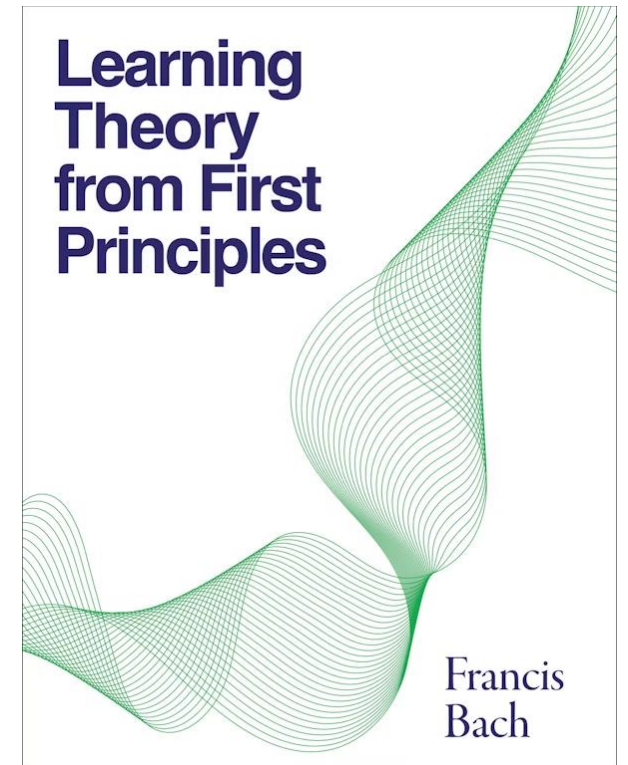
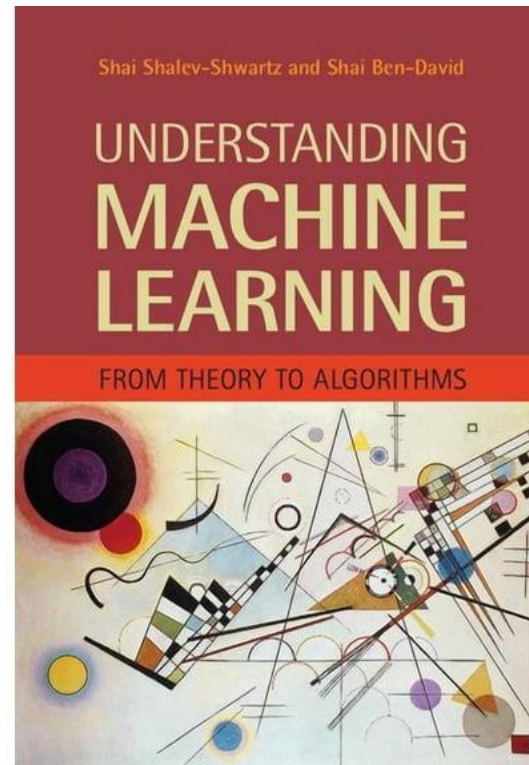
Lecture 1 (08/26): Learning Theory

https://ruizhezhang.com/course_fall_2026.html

Outline

Intro to Learning Theory

- Statistical Learning Framework
- PAC Learning
- The VC Dimension



The Statistical Learning Framework

- **Domain set $\mathcal{X}$:** images, sounds, text, DNA sequences, web pages, ...
- **Label set $\mathcal{Y}$:** for binary classification, $\mathcal{Y} = \{0,1\}$ or $\{-1,1\}$; for multcategory classification, $\mathcal{Y} = \{1,2, \dots, k\}$; and for classical regression, $\mathcal{Y} = \mathbb{R}$
- **Training data $S = ((x_1, y_1), \dots, (x_m, y_m)) \in (\mathcal{X} \times \mathcal{Y})^m$**
- **Data-generation model:** a probability distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$
 - Note that y_i may not be a deterministic function of x_i . E.g., $y = f(x) + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$, or $y = f(x, z)$ with unobserved z
 - $\mathcal{D}$ is unknown to the learner
- **Machine learning algorithm A :** a function that maps the training data S to a function (prediction rule/predictor/classifier/hypothesis) from $\mathcal{X}$ to $\mathcal{Y}$



The Statistical Learning Framework

- Loss function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$
 - 0-1 loss $\ell(y, z) = \mathbf{1}\{y \neq z\}$
 - square loss $\ell(y, z) = (y - z)^2$
- Expected risk / Population loss / Generalization error / Test error of f :

$$\mathcal{R}(f) := \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(f(x), y)]$$

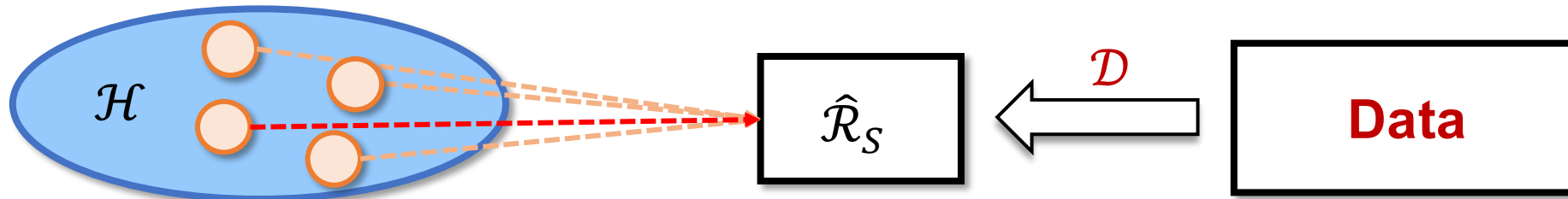
- Empirical risk / Empirical loss / Training error of f :

$$\hat{\mathcal{R}}_S(f) := \frac{1}{m} \sum_{i=1}^m \ell(f(x_i), y_i)$$

Empirical Risk Minimization

- Let $\mathcal{H}$ be a hypothesis class $\{f : \mathcal{X} \rightarrow \mathcal{Y}\}$
- An ERM learner aims at finding a function $f \in \mathcal{H}$ that minimizes the empirical risk:

$$\text{ERM}_{\mathcal{H}}(S) \in \arg \min_{f \in \mathcal{H}} \hat{R}_S(f)$$



Example: ordinary linear regression

- $\mathcal{H} = \{f_{\theta}(x) = \theta^{\top} \phi(x)\}$ where $\phi(\cdot)$ is a fixed feature map $\mathcal{X} \rightarrow \mathbb{R}^d$
- ERM is the least squares minimization:

$$\min_{\theta} \frac{1}{m} \sum_{i=1}^m (\theta^{\top} \phi(x_i) - y_i)^2$$

Empirical Risk Minimization

The Bayes optimal predictor:

$$\begin{aligned}\mathcal{R}(f) &= \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(f(x), y)] = \mathbb{E}[\mathbb{E}[\ell(f(x), y)|x]] \\ &= \int_{\mathcal{X}} \underbrace{\mathbb{E}[\ell(f(x'), y)|x = x']}_{\text{conditional risk}} dp(x'),\end{aligned}$$

where p is the density of the marginal distribution of $\mathcal{D}$ on $\mathcal{X}$

- To minimize the expected risk $\mathcal{R}(f)$, we just need to minimize the conditional risk for each $x' \in \mathcal{X}$

$$f_{\star}(x') := \arg \min_{z \in \mathcal{Y}} \mathbb{E}[\ell(z, y)|x = x'], \quad \forall x' \in \mathcal{X}$$

- The Bayes risk is

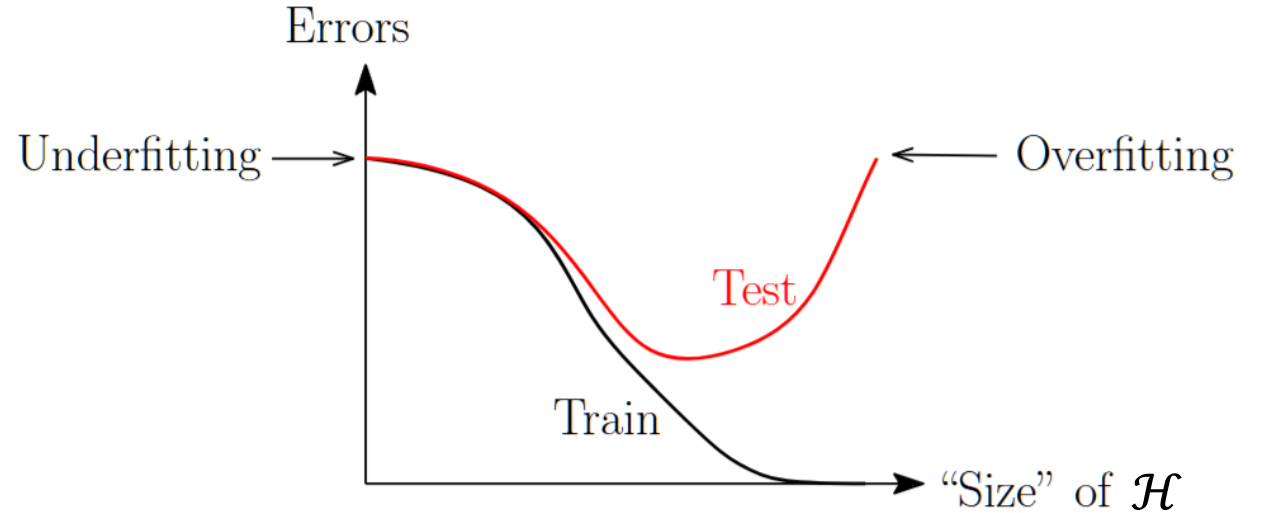
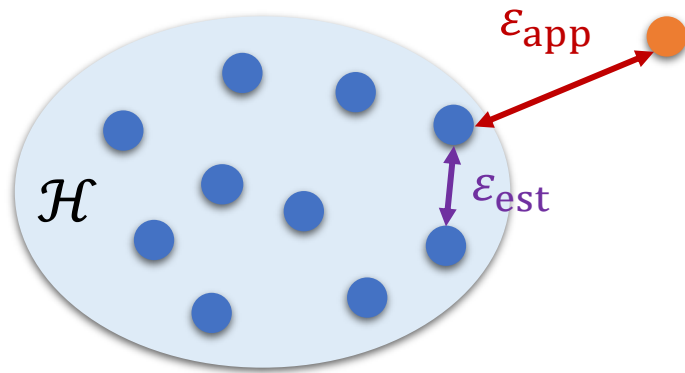
$$\mathcal{R}_{\star} := \mathcal{R}(f_{\star}) = \int_{\mathcal{X}} \min_{z \in \mathcal{Y}} \mathbb{E}[\ell(z, y)|x = x'] dp(x')$$

- **Exercise:** Show that the Bayes optimal predictor $f_{\star}$ is optimal, i.e., $\mathcal{R}_{\star} \leq \mathcal{R}(f)$ for any $f: \mathcal{X} \rightarrow \mathcal{Y}$

Empirical Risk Minimization

Risk decomposition:

$$R(f) - \mathcal{R}_\star = \underbrace{\min_{h \in \mathcal{H}} R(h) - \mathcal{R}_\star}_{\text{approximation error}} + \underbrace{R(f) - \min_{h \in \mathcal{H}} R(h)}_{\text{estimation error}}$$



Toy Example: Polynomial Regression

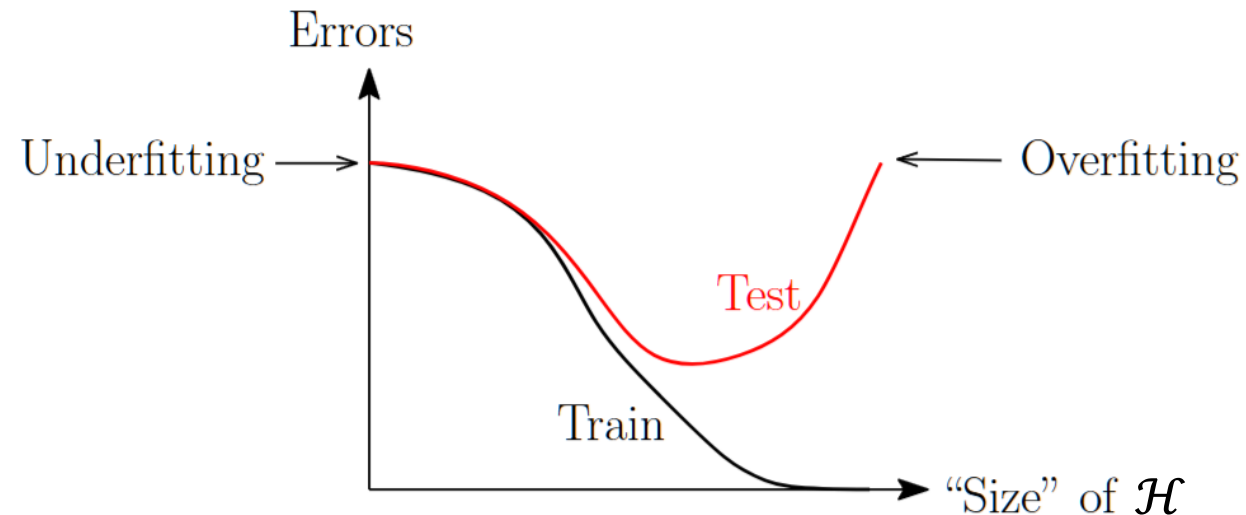
The goal is to find the best degree- P polynomial approximation of the function

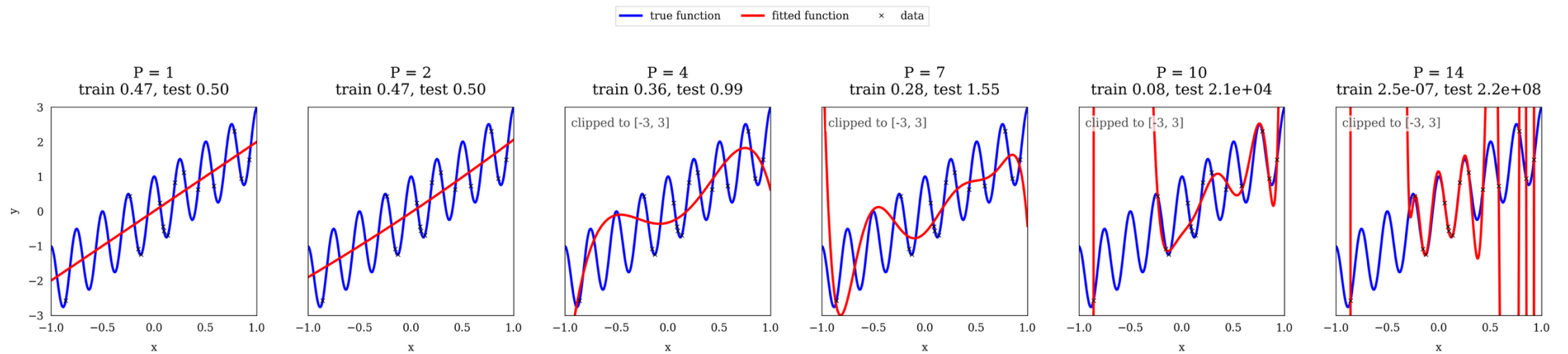
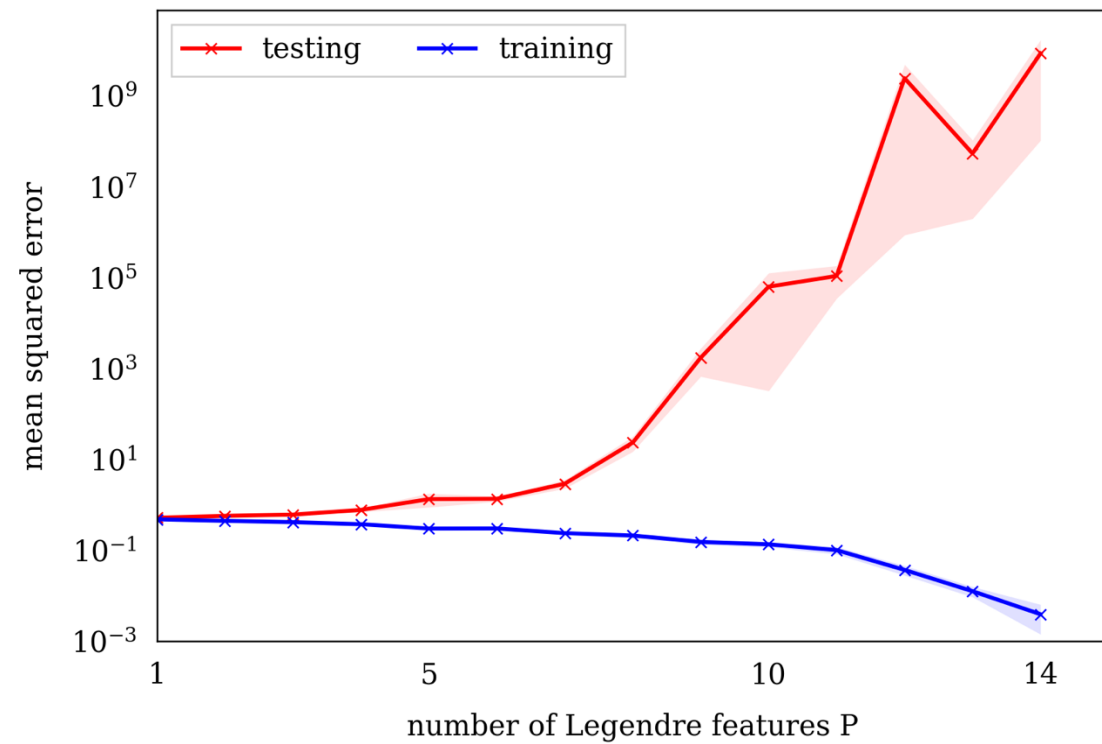
$$f(x) = 2x + \cos(25x)$$

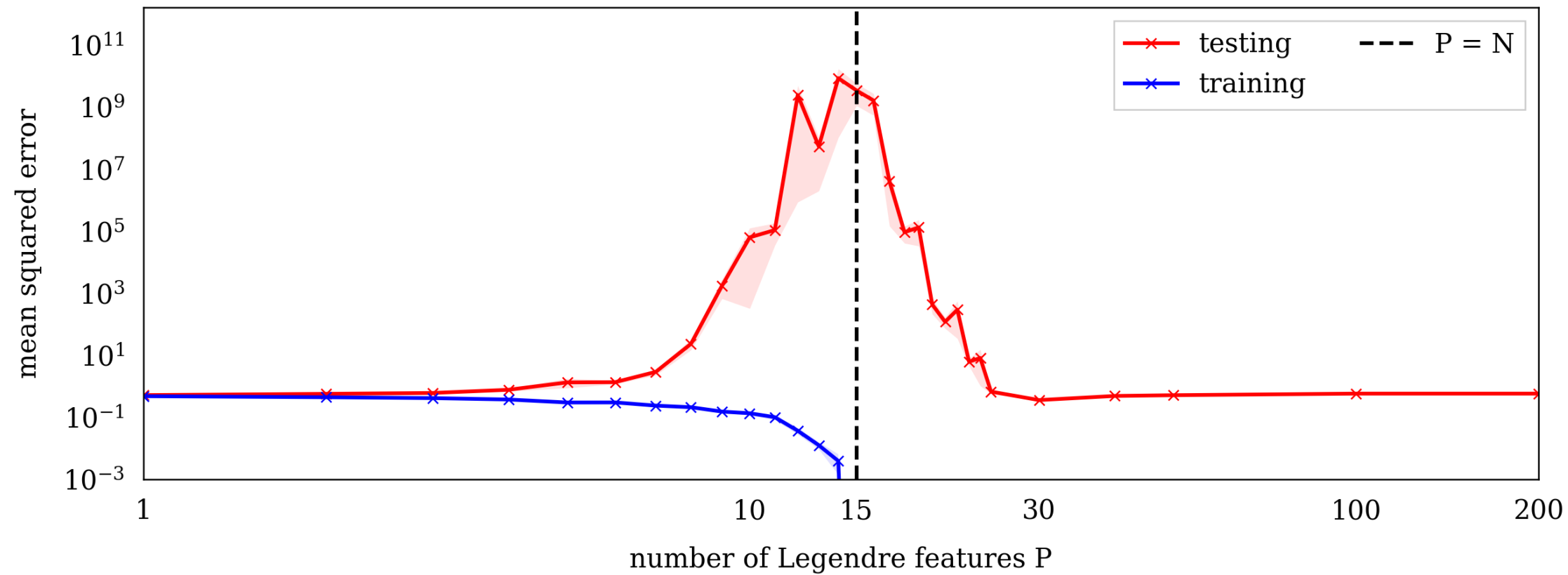
- Training data: $\{(x_i, f(x_i))\}_{i \leq N}$, each $x_i \in [-1, 1]$ is an i.i.d. uniform sample
- Hypothesis class:

$$\mathcal{H} = \{\theta \in \mathbb{R}^P : \theta^\top \phi(x)\}, \quad \phi(x) = (L_1(x), L_2(x), \dots, L_P(x))$$

Legendre polynomials







(Schaeffer et al., arXiv:2303.14151)

PAC Learning

Probably approximately correct (PAC) learning

A hypothesis class $\mathcal{H}$ is **(agnostic) PAC learnable** if there exist a function $m_{\mathcal{H}}$ and a learning algorithm A such that for any $\delta \in (0, 1)$ and $\varepsilon > 0$, for any distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$, when running the learning algorithm on $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$ i.i.d. samples S generated by $\mathcal{D}$, the algorithm returns a hypothesis $A(S) : \mathcal{X} \rightarrow \mathcal{Y}$ satisfying

$$\Pr_{S \sim \mathcal{D}^m} \left[\mathcal{R}(A(S)) - \inf_{f \in \mathcal{H}} \mathcal{R}(f) \leq \varepsilon \right] \geq 1 - \delta$$

- Today's PAC learning definition is slightly different from the original one; see [\(Hanneke-Mehrotra-Velegkas-Zampetakis, arXiv:2605.13840\)](#)
- PAC learning has strong connections to many subfields of theoretical computer science apart from learning, including property testing, algorithmic game theory, computational complexity, statistical algorithms, and cryptography; see [\(Servedio, arXiv:2511.08791\)](#)



Leslie Valiant
Turing Award Laureate

PAC Learning

Given a hypothesis class $\mathcal{H}$, is it PAC learnable? How many samples do we need?

The Realizability Assumption: there exists an $h \in \mathcal{H}$ such that for any $(x, y) \sim \mathcal{D}$ satisfies $y = h(x)$. In other words, $\mathcal{R}(h) = 0$

Proposition. Every **finite** hypothesis class is PAC learnable with sampling complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon}\right)$$

- There exists a learner A such that for any $m \geq \varepsilon^{-1} \log(|\mathcal{H}|/\delta)$,

$$\Pr_{S \sim \mathcal{D}^m}[\mathcal{R}(A(S)) \leq \varepsilon] \geq 1 - \delta$$

- With high probability over $S \sim \mathcal{D}^m$, the empirical risk minimization satisfies

Generalization bound

$$\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \underbrace{\hat{\mathcal{R}}_S(\text{ERM}_{\mathcal{H}}(S))}_{= 0} \leq O\left(\frac{\log(|\mathcal{H}|)}{m}\right)$$

PAC Learning

Proposition. Every finite hypothesis class is PAC learnable with sampling complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon}\right)$$

Proof.

- It suffices to prove that, for $m \geq \varepsilon^{-1} \log(|\mathcal{H}|/\delta)$, $\Pr_{S \sim \mathcal{D}^m}[\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) > \varepsilon] \leq \delta$

- Let $\mathcal{H}_B := \{f \in \mathcal{H} : \mathcal{R}(f) > \varepsilon\}$ be the set of “bad” hypothesis. That is,

$$\Pr_{S \sim \mathcal{D}^m}[\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) > \varepsilon] = \Pr_{S \sim \mathcal{D}^m}[\text{ERM}_{\mathcal{H}}(S) \in \mathcal{H}_B]$$

- Let $\mathcal{M} := \{S : \exists f \in \mathcal{H}_B, \hat{\mathcal{R}}_S(f) = 0\}$ be the set of “misleading” datasets. Then, we have

$$\Pr_{S \sim \mathcal{D}^m}[\text{ERM}_{\mathcal{H}}(S) \in \mathcal{H}_B] \leq \Pr_{S \sim \mathcal{D}^m}[S \in \mathcal{M}] = \Pr_{S \sim \mathcal{D}^m}[\exists f \in \mathcal{H}_B, \hat{\mathcal{R}}_S(f) = 0] \leq \sum_{f \in \mathcal{H}_B} \Pr_{S \sim \mathcal{D}^m}[\hat{\mathcal{R}}_S(f) = 0]$$

(Union bound)

PAC Learning

Proposition. Every finite hypothesis class is PAC learnable with sampling complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon}\right)$$

Proof.

$$\begin{aligned} \Pr_{S \sim \mathcal{D}^m}[\text{ERM}_{\mathcal{H}}(S) \in \mathcal{H}_B] &\leq \sum_{f \in \mathcal{H}_B} \Pr_{S \sim \mathcal{D}^m}[\hat{\mathcal{R}}_S(f) = 0] = \sum_{f \in \mathcal{H}_B} \Pr_{S \sim \mathcal{D}^m}[\forall i \in [m], f(x_i) = y_i] \\ &= \sum_{f \in \mathcal{H}_B} \left(\Pr_{(x,y) \sim \mathcal{D}}[f(x) = y] \right)^m \\ &\leq \sum_{f \in \mathcal{H}_B} \left(\mathbb{E}_{(x,y) \sim \mathcal{D}}[1 - \ell(f(x), y)] \right)^m \quad \ell \text{ is the 0-1 loss or bounded loss} \\ &= \sum_{f \in \mathcal{H}_B} (1 - \mathcal{R}(f))^m \leq \sum_{f \in \mathcal{H}_B} (1 - \varepsilon)^m \leq |\mathcal{H}_B| e^{-\varepsilon m} \leq \delta \end{aligned}$$

■

PAC Learning

“No Free Lunch” Theorem

Consider binary classification with 0–1 loss and $\mathcal{X}$ infinite. For any $m > 0$ and any learning algorithm A , there exists a probability distribution $\mathcal{D}$ such that

$$\mathcal{R}_\star = 0 \quad \text{and} \quad \mathbb{E}_{S \sim \mathcal{D}^m} [\mathcal{R}(A(S))] \geq 1/4$$

- **No universal learner:** for every learner, there exists a task on which it fails, even though that task can be successfully learned by another learner
- **Exercise:** Let $\mathcal{X}$ be infinite and let $\mathcal{H}$ be the set of **all functions** from $\mathcal{X}$ to $\{0,1\}$. Then, $\mathcal{H}$ is not PAC learnable

PAC Learning

“No Free Lunch” Theorem Consider binary classification with 0–1 loss and $\mathcal{X}$ infinite. For any $m > 0$ and any learning algorithm A , there exists a probability distribution $\mathcal{D}$ such that

$$\mathcal{R}_\star = 0 \quad \text{and} \quad \mathbb{E}_{S \sim \mathcal{D}^m} [\mathcal{R}(A(S))] \geq 1/4$$

Proof.

- Fix any $m > 0$. The idea is to construct a hard distribution $\mathcal{D}$ supported on $k \gg m$ data points such that the knowledge of the m training data point is not enough to predict the remaining $k - m$ data points
- Pick any k different elements $x_1, \dots, x_k \in \mathcal{X}$
- For every $r \in \{0,1\}^k$, define a probability distribution $\mathcal{D}_r$:

$$\mathcal{D}_r((x_i, y)) := \begin{cases} 1/k & \text{if } y = r_i \\ 0 & \text{otherwise} \end{cases}, \quad i \in [k], y \in \{0,1\}$$

- $\mathcal{R}_\star = 0$ for every $\mathcal{D}_r$

PAC Learning

“No Free Lunch” Theorem Consider binary classification with 0–1 loss and $\mathcal{X}$ infinite. For any $m > 0$ and any learning algorithm A , there exists a probability distribution $\mathcal{D}$ such that

$$\mathcal{R}_\star = 0 \quad \text{and} \quad \mathbb{E}_{S \sim \mathcal{D}^m} [\mathcal{R}(A(S))] \geq 1/4$$

Proof.

- Probabilistic method:

$$\begin{aligned} \max_{r \in \{0,1\}^k} \{ \mathbb{E}_{S \sim \mathcal{D}_r^m} [\mathcal{R}(A(S))] \} &\geq \mathbb{E}_{r \sim \{0,1\}^k} \left[\mathbb{E}_{S \sim \mathcal{D}_r^m} [\mathcal{R}(A(S))] \right] \\ &= \mathbb{E}_{r,S} \left[\Pr_{i \sim [k]} [A(S)(x_i) \neq r_i] \right] = \Pr_{r,S,i} [A(S)(x_i) \neq r_i] \end{aligned}$$

- To prove the existence of some object with certain property, we devise a *random construction* and show that it works with *positive probability*. (Here, we use $\Pr[X \geq \mathbb{E}[X]] > 0$)

PAC Learning

“No Free Lunch” Theorem Consider binary classification with 0–1 loss and $\mathcal{X}$ infinite. For any $m > 0$ and any learning algorithm A , there exists a probability distribution $\mathcal{D}$ such that

$$\mathcal{R}_\star = 0 \quad \text{and} \quad \mathbb{E}_{S \sim \mathcal{D}^m}[\mathcal{R}(A(S))] \geq 1/4$$

Proof.

- Let $S = (x_{i_1}, r_{i_1}, \dots, x_{i_m}, r_{i_m})$

$$\Pr_{r, S, i}[A(S)(x_i) \neq r_i] = \mathbb{E} \left[\Pr_{r, i}[A(S)(x_i) \neq r_i \mid x_{i_1}, r_{i_1}, \dots, x_{i_m}, r_{i_m}] \right]$$

PAC Learning

“No Free Lunch” Theorem Consider binary classification with 0–1 loss and $\mathcal{X}$ infinite. For any $m > 0$ and any learning algorithm A , there exists a probability distribution $\mathcal{D}$ such that

$$\mathcal{R}_\star = 0 \quad \text{and} \quad \mathbb{E}_{S \sim \mathcal{D}^m}[\mathcal{R}(A(S))] \geq 1/4$$

Proof.

- Thus, we get that

$$\max_{r \in \{0,1\}^k} \{\mathbb{E}_{S \sim \mathcal{D}_r^m}[\mathcal{R}(A(S))]\} \geq \frac{1}{2} \left(1 - \frac{m}{k}\right) \geq \frac{1}{4}$$

when $k \geq 2m$



PAC Learning

- We have shown that under the **realizability assumption** ($\exists h \in \mathcal{H}, \mathcal{R}(h) = 0$), every finite $\mathcal{H}$ is PAC learnable
- We can remove this assumption:

Theorem. Let $\mathcal{H}$ be a finite class, and ℓ be the loss function bounded in $[0,1]$. Then, $\mathcal{H}$ is **agnostic** PAC learnable using the ERM algorithm with sample complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon^2}\right)$$

- A byproduct is the **Uniform Convergence property**: for any $\mathcal{D}$, with high probability over $S \sim \mathcal{D}^m$,

$$\max_{f \in \mathcal{H}} |\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \leq O\left(\sqrt{\frac{\log(|\mathcal{H}|)}{m}}\right)$$

PAC Learning

Theorem. Let $\mathcal{H}$ be a finite class, and ℓ be the loss function bounded in $[0,1]$. Then, $\mathcal{H}$ is agnostically PAC learnable using the ERM algorithm with sample complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon^2}\right)$$

Proof.

- First, we want to prove that for $m \geq m(\varepsilon, \delta)$,

$$\Pr_{S \sim \mathcal{D}^m} \left[\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \inf_{f \in \mathcal{H}} \mathcal{R}(f) \geq \varepsilon \right] \leq \delta$$

- Let $h := \arg \min_{f \in \mathcal{H}} \mathcal{R}(f)$. We have the following error decomposition:

$$\begin{aligned} & \mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \mathcal{R}(h) \\ &= \left(\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \hat{\mathcal{R}}_S(\text{ERM}_{\mathcal{H}}(S)) \right) + \overbrace{\left(\hat{\mathcal{R}}_S(\text{ERM}_{\mathcal{H}}(S)) - \hat{\mathcal{R}}_S(h) \right)}^{\leq 0} + \left(\hat{\mathcal{R}}_S(h) - \mathcal{R}(h) \right) \\ &\geq \left(\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \hat{\mathcal{R}}_S(\text{ERM}_{\mathcal{H}}(S)) \right) + \left(\hat{\mathcal{R}}_S(h) - \mathcal{R}(h) \right) \end{aligned}$$

PAC Learning

Theorem. Let $\mathcal{H}$ be a finite class, and ℓ be the loss function bounded in $[0,1]$. Then, $\mathcal{H}$ is agnostically PAC learnable using the ERM algorithm with sample complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon^2}\right)$$

Proof.

$$\begin{aligned}\Pr_{S \sim \mathcal{D}^m}[\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \mathcal{R}(h) \geq \varepsilon] &\leq \Pr_{S \sim \mathcal{D}^m}\left[\left(\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \hat{\mathcal{R}}_S(\text{ERM}_{\mathcal{H}}(S))\right) + \left(\hat{\mathcal{R}}_S(h) - \mathcal{R}(h)\right) \geq \varepsilon\right] \\ &\leq \Pr_{S \sim \mathcal{D}^m}[\exists f \in \mathcal{H}, |\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \geq \varepsilon/2] \\ &\leq \sum_{f \in \mathcal{H}} \Pr_{S \sim \mathcal{D}^m}[|\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \geq \varepsilon/2]\end{aligned}$$

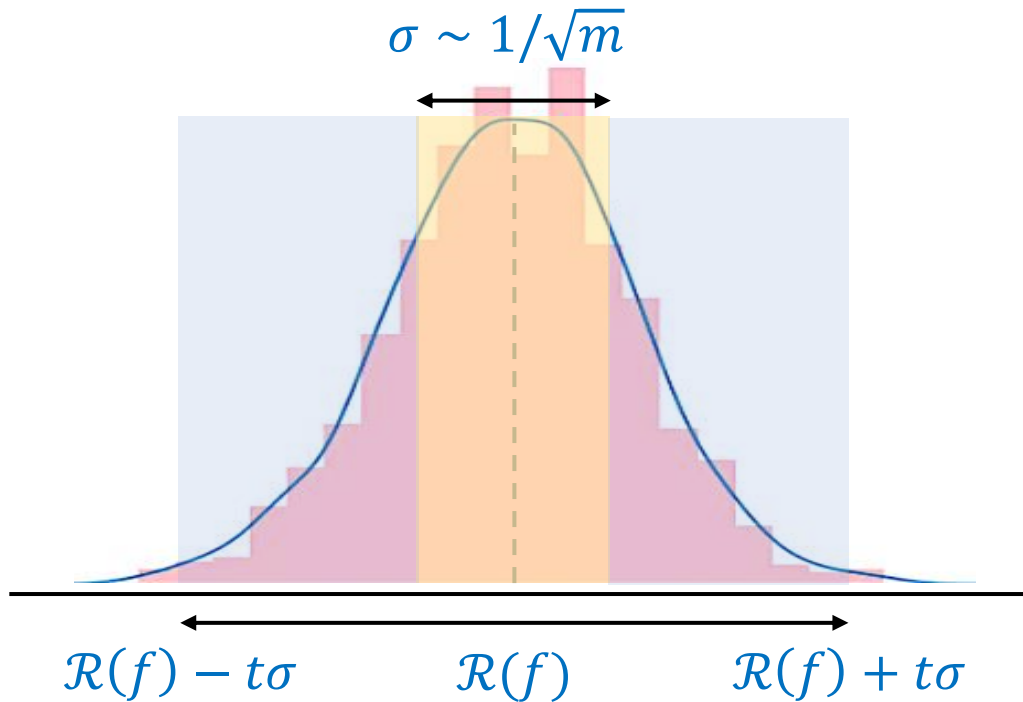
Random variable

- Note that

$$\hat{\mathcal{R}}_S(f) = \frac{1}{m} \sum_{i=1}^m \ell(f(x_i), y_i), \quad \mathcal{R}(f) = \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(f(x), y)], \quad \mathbb{E}[\hat{\mathcal{R}}(f)] = \mathcal{R}(f)$$

Detour: Concentration

Central limit theorem / Hoeffding / Chernoff / ...



$$\Pr[|X - \mathbb{E}[X]| \geq t\sigma] \leq \exp(-\Omega(t^2))$$

Hoeffding's inequality

Let $X_1, \dots, X_m$ be i.i.d. random variables with $\mathbb{E}[X_i] = \mu$ and $a \leq X_i \leq b$. Then,

$$\Pr\left[\left|\frac{1}{m} \sum_{i=1}^m X_i - \mu\right| \geq \epsilon\right] \leq 2 \exp\left(-\frac{2m\epsilon^2}{(b-a)^2}\right)$$

https://ruizhezhang.com/teaching_2026/Lecture%203.pdf

PAC Learning

Theorem. Let $\mathcal{H}$ be a finite class, and ℓ be the loss function bounded in $[0,1]$. Then, $\mathcal{H}$ is agnostically PAC learnable using the ERM algorithm with sample complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon^2}\right)$$

Proof.

- By Hoeffding's inequality,

$$\Pr_{S \sim \mathcal{D}^m} [|\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \geq \varepsilon/2] \leq \exp(-\Omega(m\varepsilon^2))$$

- Thus,

$$\Pr_{S \sim \mathcal{D}^m} \left[\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \inf_{f \in \mathcal{H}} \mathcal{R}(f) \geq \varepsilon \right] \leq |\mathcal{H}| \cdot \exp(-\Omega(m\varepsilon^2)) \leq \delta$$

as long as

$$m \geq \Omega\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon^2}\right)$$



PAC Learning

Theorem. Let $\mathcal{H}$ be a finite class, and ℓ be the loss function bounded in $[0,1]$. Then, $\mathcal{H}$ is agnostically PAC learnable using the ERM algorithm with sample complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon^2}\right)$$

What about infinite class $|\mathcal{H}| = \infty$?

- **No Free Lunch theorem** tells that $\mathcal{H}$ cannot be all functions from an infinite domain $\mathcal{X}$ to $\{0,1\}$
- **Exercise:** Let $\mathcal{X} = \mathbb{R}$ and $\mathcal{H}$ be the set of indicator functions of closed intervals:

$$\mathbf{1}\{a \leq x \leq b\}, \quad \forall a \leq b$$

Then $\mathcal{H}$ can be PAC-learned using ERM with sample complexity $O\left(\frac{1}{\varepsilon} \log \frac{1}{\delta}\right)$

The VC Dimension

Shattering: A hypothesis class $\mathcal{H}$ shatters a finite set $C \subset \mathcal{X}$ if the restriction of $\mathcal{H}$ to C is the set of all functions from C to $\{0, 1\}$, i.e., for $C = \{c_1, \dots, c_m\}$,

$$\mathcal{H}_C := \{(h(c_1), \dots, h(c_m)) : h \in \mathcal{H}\} = \{(y_1, \dots, y_m) : y_i \in \{0, 1\}\}$$

That is, $|\mathcal{H}_C| = 2^{|C|}$

	c_1	c_2	c_3	c_4	c_5
h_1	0	0	0	0	0
h_2	1	1	1	1	1
h_3	1	1	1	0	0
h_4	0	0	0	1	1

The VC Dimension

Shattering: A hypothesis class $\mathcal{H}$ shatters a finite set $C \subset \mathcal{X}$ if the restriction of $\mathcal{H}$ to C is the set of all functions from C to $\{0, 1\}$, i.e., for $C = \{c_1, \dots, c_m\}$,

$$\mathcal{H}_C := \{(h(c_1), \dots, h(c_m)) : h \in \mathcal{H}\} = \{(y_1, \dots, y_m) : y_i \in \{0, 1\}\}$$

That is, $|\mathcal{H}_C| = 2^{|C|}$

VC-dimension

The VC-dimension of a hypothesis class $\mathcal{H}$, denoted $\text{VCdim}(\mathcal{H})$, is the maximal size of a set $C \subset \mathcal{X}$ that can be shattered by $\mathcal{H}$. If $\mathcal{H}$ can shatter sets of arbitrarily large size, we say $\text{VCdim}(\mathcal{H}) = \infty$

- If $\text{VCdim}(\mathcal{H}) = \infty$, then $\mathcal{H}$ is not PAC learnable



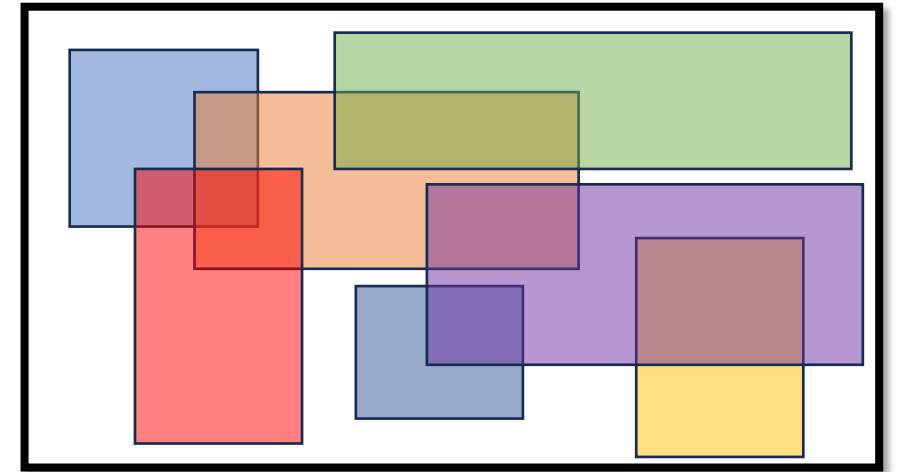
Alexei Chervonenkis and Vladimir Vapnik

The VC Dimension

The VC-dimension of a hypothesis class $\mathcal{H}$, denoted $\text{VCdim}(\mathcal{H})$, is the **maximal** size of a set $C \subset \mathcal{X}$ that can be shattered by $\mathcal{H}$.

Examples:

- Finite $\mathcal{H}$: $2^{|C|} = |\mathcal{H}_C| \leq |\mathcal{H}| \Rightarrow \text{VCdim}(\mathcal{H}) \leq \log|\mathcal{H}|$
- Axis aligned rectangles: $\text{VCdim}(\mathcal{H}) = 4$



The VC Dimension

The Fundamental Theorem of Statistical Learning.

Let $\mathcal{H}$ be a hypothesis class of functions from $\mathcal{X}$ to $\{0, 1\}$ and let the loss function be the 0–1 loss. Then, the following are equivalent:

- 1) $\mathcal{H}$ is agnostic PAC learnable
- 2) $\mathcal{H}$ is PAC learnable
- 3) $\mathcal{H}$ has finite VC dimension

- We already have $1) \Rightarrow 2)$ and $2) \Rightarrow 3)$. It remains to show that $3) \Rightarrow 1)$
- It is very similar to the agnostic PAC learning theorem for finite $\mathcal{H}$:

Let $\mathcal{H}$ be a finite class. Then, $\mathcal{H}$ is agnostic PAC learnable with sample complexity $O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon^2}\right)$

- The VC dimension bounds the “effective size” of $\mathcal{H}$

The VC Dimension

Lemma (Sauer-Shelah-Perles). Let $\mathcal{H}$ be a hypothesis class with $\text{VCdim}(\mathcal{H}) \leq d < \infty$. Then, for all m ,

$$\tau_{\mathcal{H}}(m) := \max_{C \subset \mathcal{X}: |C| \leq m} |\mathcal{H}_C| \leq \sum_{i=0}^d \binom{m}{i}$$

In particular, if $m \geq d + 1$, then $\tau_{\mathcal{H}}(m) \leq (em/d)^d$

- For any subset C , the “effective size” $|\mathcal{H}_C| = O(|C|^d)$ instead of $2^{|C|}$
- We only prove the uniform convergence property, which will give agnostic PAC learnability

Proposition (Uniform Convergence). Let $\mathcal{H}$ be a class and let $\tau_{\mathcal{H}}$ be its growth function. Then, for every $\mathcal{D}$, with probability of at least $1 - \delta$ over the choice of $S \sim \mathcal{D}^m$ we have

$$\max_{f \in \mathcal{H}} |\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \leq \frac{\sqrt{\log \tau_{\mathcal{H}}(2m)}}{\delta \sqrt{m}}$$

The VC Dimension

Proposition. Let $\mathcal{H}$ be a class and let $\tau_{\mathcal{H}}$ be its growth function. Then, for every $\mathcal{D}$, with probability of at least $1 - \delta$ over the choice of $S \sim \mathcal{D}^m$ we have

$$\max_{f \in \mathcal{H}} |\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \leq O\left(\frac{\sqrt{1 + \log \tau_{\mathcal{H}}(2m)}}{\delta \sqrt{m}}\right)$$

Proof.

- It suffices to prove that $\mathbb{E}_S \left[\max_{f \in \mathcal{H}} |\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \right] \leq O\left(\sqrt{\frac{1 + \log \tau_{\mathcal{H}}(2m)}{m}}\right)$
- Recall that $\mathbb{E}_S[\hat{\mathcal{R}}_S(f)] = \mathcal{R}(f)$. We can use the reverse direction:

$$\begin{aligned} \mathbb{E}_S \left[\max_{f \in \mathcal{H}} |\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \right] &= \mathbb{E}_S \left[\max_{f \in \mathcal{H}} |\mathbb{E}_{S'}[\hat{\mathcal{R}}_{S'}(f)] - \hat{\mathcal{R}}_S(f)| \right] \\ &\leq \mathbb{E}_S \left[\max_{f \in \mathcal{H}} \{ \mathbb{E}_{S'}[|\hat{\mathcal{R}}_{S'}(f) - \hat{\mathcal{R}}_S(f)|] \} \right] \end{aligned}$$

The VC Dimension

- By Jensen's inequality,

$$\max_{f \in \mathcal{H}} \{ \mathbb{E}_{S'} [| \hat{\mathcal{R}}_{S'}(f) - \hat{\mathcal{R}}_S(f) |] \} \leq \mathbb{E}_{S'} \left[\max_{f \in \mathcal{H}} | \hat{\mathcal{R}}_{S'}(f) - \hat{\mathcal{R}}_S(f) | \right]$$

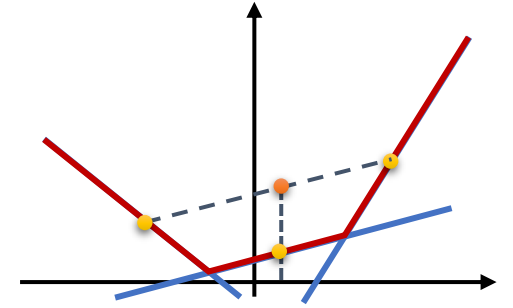
- Thus,

$$\mathbb{E}_S \left[\max_{f \in \mathcal{H}} | \mathcal{R}(f) - \hat{\mathcal{R}}_S(f) | \right] \leq \mathbb{E}_{S,S'} \left[\max_{f \in \mathcal{H}} | \hat{\mathcal{R}}_{S'}(f) - \hat{\mathcal{R}}_S(f) | \right]$$

- Fix $S, S' \sim \mathcal{D}^m$, and let $\mathcal{C} := S \cup S'$. Then,

$$\max_{f \in \mathcal{H}} | \hat{\mathcal{R}}_{S'}(f) - \hat{\mathcal{R}}_S(f) | = \max_{f \in \mathcal{H}_{\mathcal{C}}} | \hat{\mathcal{R}}_{S'}(f) - \hat{\mathcal{R}}_S(f) |$$

$$\begin{aligned} \mathbb{E}_S \left[\max_{f \in \mathcal{H}} | \mathcal{R}(f) - \hat{\mathcal{R}}_S(f) | \right] &\leq \mathbb{E}_{S,S'} \left[\max_{f \in \mathcal{H}_{\mathcal{C}}} | \hat{\mathcal{R}}_{S'}(f) - \hat{\mathcal{R}}_S(f) | \right] \\ &= \mathbb{E}_{S,S'} \left[\max_{f \in \mathcal{H}_{\mathcal{C}}} \left| \frac{1}{m} \sum_{i=1}^m \ell(f(x'_i), y'_i) - \ell(f(x_i), y_i) \right| \right] \end{aligned}$$



The VC Dimension

- Note the symmetry between S and S' : for any $\sigma \in \{\pm 1\}^m$,

$$\mathbb{E}_{S,S'} \left[\max_{f \in \mathcal{H}_C} \left| \frac{1}{m} \sum_{i=1}^m \ell(f(x'_i), y'_i) - \ell(f(x_i), y_i) \right| \right] = \mathbb{E}_{S,S'} \left[\max_{f \in \mathcal{H}_C} \left| \frac{1}{m} \sum_{i=1}^m \sigma_i (\ell(f(x'_i), y'_i) - \ell(f(x_i), y_i)) \right| \right]$$

- Thus, we can take expectation over a uniformly random $\sigma \sim \{\pm 1\}^n$:

$$\text{LHS} = \mathbb{E}_{S,S'} \mathbb{E}_{\sigma} \left[\max_{f \in \mathcal{H}_C} \left| \frac{1}{m} \sum_{i=1}^m \sigma_i (\ell(f(x'_i), y'_i) - \ell(f(x_i), y_i)) \right| \right]$$

- $\sigma_i (\ell(f(x'_i), y'_i) - \ell(f(x_i), y_i))$ is a mean-zero random variable in $\{-1, 0, 1\}$. By Hoeffding's inequality,

$$\Pr_{\sigma} \left[\left| \frac{1}{m} \sum_{i=1}^m \sigma_i (\ell(f(x'_i), y'_i) - \ell(f(x_i), y_i)) \right| \geq \epsilon \right] \leq \exp(-\Omega(m\epsilon^2))$$

The VC Dimension

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The VC Dimension

- Using the fact that for any nonnegative random variable X ,

$$\mathbb{E}[X] = \int_0^\infty \Pr[X \geq t] dt,$$

we have

$$\begin{aligned} & \mathbb{E}_\sigma \left[\max_{f \in \mathcal{H}_C} \left| \frac{1}{m} \sum_{i=1}^m \sigma_i (\ell(f(x'_i), y'_i) - \ell(f(x_i), y_i)) \right| \right] \\ &= \int_0^\infty \Pr_\sigma \left[\max_{f \in \mathcal{H}_C} \left| \frac{1}{m} \sum_{i=1}^m \sigma_i (\ell(f(x'_i), y'_i) - \ell(f(x_i), y_i)) \right| \geq \epsilon \right] d\epsilon \\ &\leq \int_0^\infty \min\{1, \tau_{\mathcal{H}}(2m) \exp(-\Omega(m\epsilon^2))\} d\epsilon \\ &= O\left(\frac{\sqrt{1 + \log \tau_{\mathcal{H}}(2m)}}{\sqrt{m}}\right) \end{aligned}$$



The VC Dimension

$$\int_0^\infty \min\{1, \tau_{\mathcal{H}}(2m) \exp(-\Omega(m\epsilon^2))\} d\epsilon$$

Worked for 1m 55s

Let

$$A = \tau_{\mathcal{H}}(2m), \quad \exp(-\Omega(m\epsilon^2)) \leq e^{-cm\epsilon^2}$$

for some constant $c > 0$. Consider

$$I = \int_0^\infty \min\{1, Ae^{-cm\epsilon^2}\} d\epsilon.$$

The two terms inside the minimum are equal when

$$Ae^{-cm\epsilon_*^2} = 1 \implies \epsilon_* = \sqrt{\frac{\log A}{cm}}.$$

Thus, for $A \geq 1$,

$$\begin{aligned} I &= \int_0^{\epsilon_*} 1 d\epsilon + A \int_{\epsilon_*}^\infty e^{-cm\epsilon^2} d\epsilon \\ &= \sqrt{\frac{\log A}{cm}} + \frac{A\sqrt{\pi}}{2\sqrt{cm}} \operatorname{erfc}\left(\sqrt{\log A}\right). \end{aligned}$$

Using the Gaussian-tail bound

$$\int_x^\infty e^{-cm t^2} dt \leq \frac{e^{-cm x^2}}{2cm x},$$

we obtain

$$I \leq \sqrt{\frac{\log A}{cm}} + \frac{1}{2\sqrt{cm \log A}}.$$

Therefore, in a form valid even when A is close to 1,

$$\int_0^\infty \min\{1, \tau_{\mathcal{H}}(2m)e^{-\Omega(m\epsilon^2)}\} d\epsilon = O\left(\sqrt{\frac{1 + \log \tau_{\mathcal{H}}(2m)}{m}}\right).$$

The VC Dimension

Lemma (Sauer-Shelah-Perles). Let $\mathcal{H}$ be a hypothesis class with $\text{VCdim}(\mathcal{H}) \leq d < \infty$. Then, for all m ,

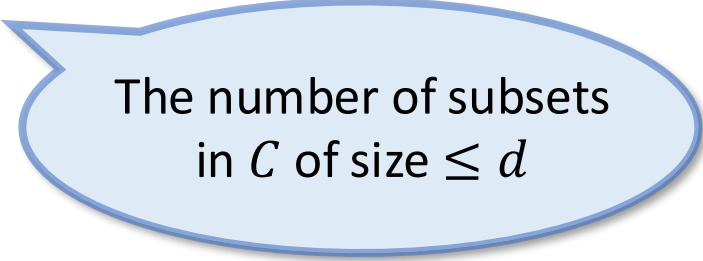
$$\tau_{\mathcal{H}}(m) := \max_{C \subset \mathcal{X} : |C| \leq m} |\mathcal{H}_C| \leq \sum_{i=0}^d \binom{m}{i}$$

Proof.

- We claim that for any $\mathcal{H}$ and $C \subset \mathcal{X}$,
$$|\mathcal{H}_C| \leq |\{B \subseteq C : \mathcal{H} \text{ shatters } B\}|$$
- We prove this claim by induction. Suppose it holds for $k < m$
- Fix $\mathcal{H}$ and $C = \{c_1, \dots, c_m\}$
- Define two sets:

$$Y_0 = \{(y_2, \dots, y_m) : (0, y_2, \dots, y_m) \in \mathcal{H}_C \wedge (1, y_2, \dots, y_m) \in \mathcal{H}_C\}$$

$$Y_1 = \{(y_2, \dots, y_m) : (0, y_2, \dots, y_m) \in \mathcal{H}_C \vee (1, y_2, \dots, y_m) \in \mathcal{H}_C\}$$



The number of subsets
in C of size $\leq d$

The VC Dimension

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$$Y_0 = \{(y_2, \dots, y_m) : (0, y_2, \dots, y_m) \in \mathcal{H}_C \wedge (1, y_2, \dots, y_m) \in \mathcal{H}_C\}$$

$$Y_1 = \{(y_2, \dots, y_m) : (0, y_2, \dots, y_m) \in \mathcal{H}_C \vee (1, y_2, \dots, y_m) \in \mathcal{H}_C\}$$

- $|Y_0| + |Y_1| = |\mathcal{H}_C|$

- Since $Y_0 = \mathcal{H}_{C'}$ with $C' := \{c_2, \dots, c_m\}$, we have

$$|Y_0| = |\mathcal{H}_{C'}| \leq |\{B \subseteq C' : \mathcal{H} \text{ shatters } B\}| = |\{B \subseteq C : c_1 \notin C \wedge \mathcal{H} \text{ shatters } B\}|$$

- Define $\mathcal{H}' \subset \mathcal{H}$ containing pairs of functions that only differ in c_1 :

$$\mathcal{H}' := \{h \in \mathcal{H} : \exists h' \in \mathcal{H} \text{ s.t. } h(c_i) = h'(c_i) \forall 2 \leq i \leq m \wedge 1 - h(c_1) = h'(c_1)\}$$

- Since $Y_1 = \mathcal{H}'_{C'}$, we have

$$\begin{aligned} |Y_1| &= |\mathcal{H}'_{C'}| \leq |\{B \subseteq C' : \mathcal{H}' \text{ shatters } B\}| = |\{B \subseteq C' : \mathcal{H}' \text{ shatters } B \cup \{c_1\}\}| \\ &= |\{B \subseteq C : c_1 \in B \wedge \mathcal{H}' \text{ shatters } B\}| \\ &\leq |\{B \subseteq C : c_1 \in B \wedge \mathcal{H} \text{ shatters } B\}| \end{aligned}$$

The VC Dimension

- Thus,

$$\begin{aligned} |\mathcal{H}_C| &= |Y_0| + |Y_1| \\ &\leq |\{B \subseteq C : c_1 \notin C \wedge \mathcal{H} \text{ shatters } B\}| + |\{B \subseteq C : c_1 \in B \wedge \mathcal{H} \text{ shatters } B\}| \\ &= |\{B \subseteq C : \mathcal{H} \text{ shatters } B\}| \end{aligned}$$



The VC Dimension

The Fundamental Theorem of Statistical Learning (quantitate version).

Let $\mathcal{H}$ be a hypothesis class of functions from $\mathcal{X}$ to $\{0, 1\}$ and let the loss function be the 0–1 loss. Suppose $\text{VCdim}(\mathcal{H}) = d$. Then

1) $\mathcal{H}$ is agnostic PAC learnable with sample complexity

$$m(\varepsilon, \delta) = \Theta\left(\frac{1 + \log(1/\delta)}{\varepsilon^2}\right)$$

2) $\mathcal{H}$ is PAC learnable with sample complexity

$$\Omega\left(\frac{d + \log(1/\delta)}{\varepsilon}\right) \leq m(\varepsilon, \delta) \leq O\left(\frac{d \log(1/\varepsilon) + \log(1/\delta)}{\varepsilon}\right)$$

Steve Hanneke (2016): $\mathcal{H}$ is **improperly** PAC learnable with sample complexity $\Theta\left(\frac{d + \log(1/\delta)}{\varepsilon}\right)$

